

Assignment - 1

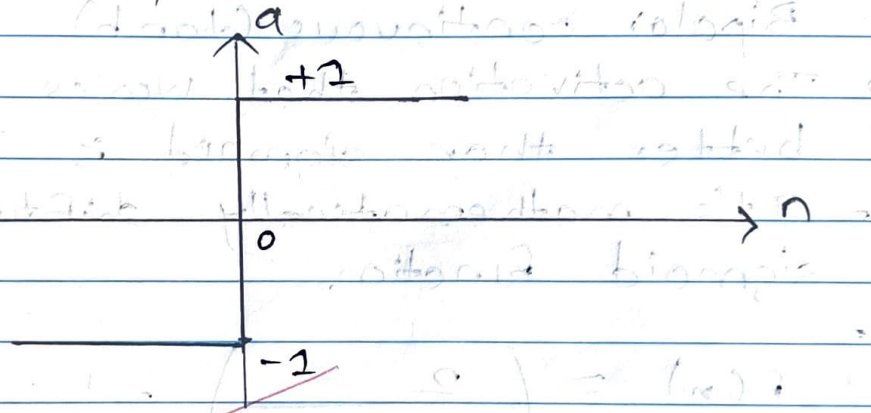
Topic - Discrete Perceptron Learning Algorithm

Q.1 Explain the following activation functions with appropriate graphs.

a) Bipolar binary (sign)

⇒ The sign function is a threshold-based activation function.

It outputs 1 for positive inputs and -1 for negative inputs.



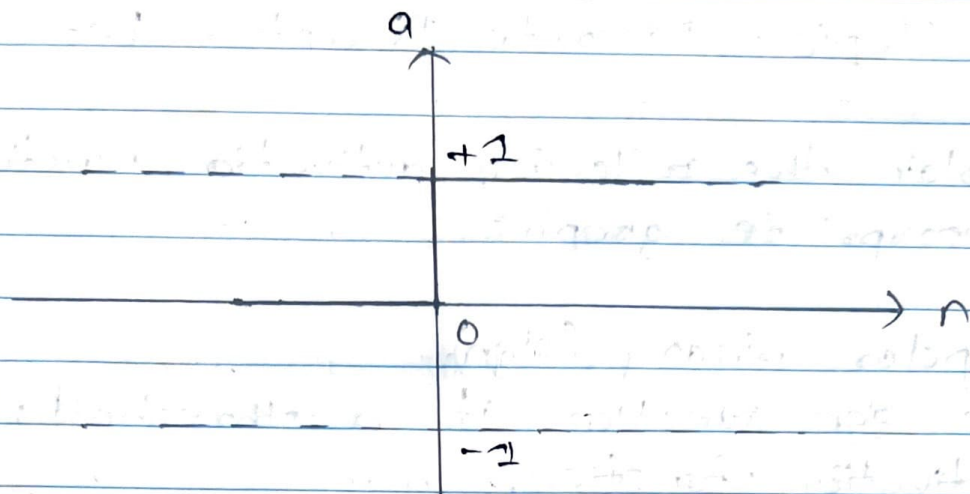
$$f(x) = \begin{cases} +1 & , x \geq 0 \\ -1 & , x < 0 \end{cases}$$

b) Unipolar Binary (step)

⇒ The step function is a binary threshold activation function.

It outputs 1 for inputs greater than or equal to 0 and 0 otherwise.

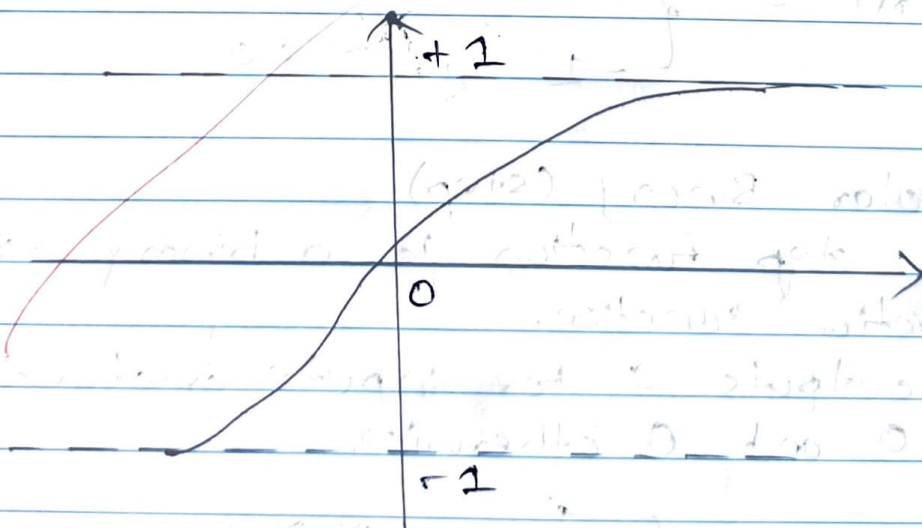
$$f(x) = \begin{cases} 1 & , x \geq 0 \\ 0 & , x < 0 \end{cases}$$



3) Bipolar continuous (tanh)

- ⇒ The activation that works almost always better than sigmoid is Tanh function.
- It's mathematically shifted version of sigmoid function.

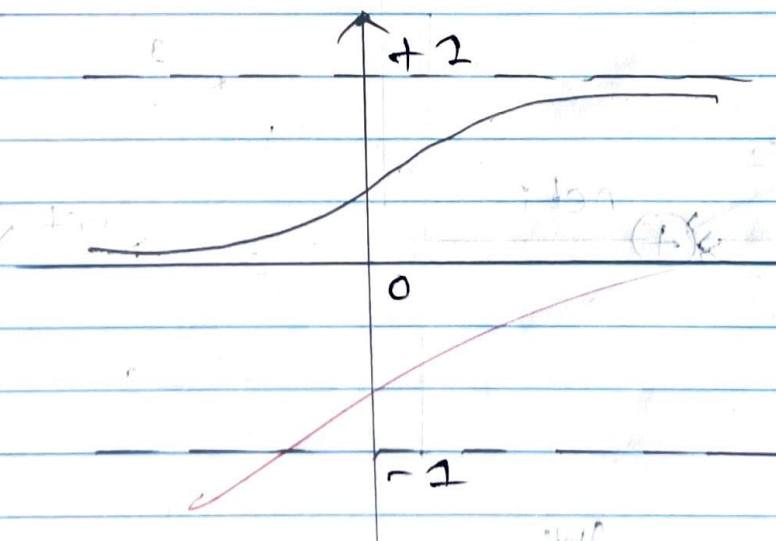
$$F(n) = \left(\frac{2}{1 + e^{-x_{net}}} \right) - 1$$



d) Unipolar continuous (sigmoid)

=) The sigmoid transfer function shown below takes the input, which can have any value between plus and minus infinity, and squashes the output into the range 0 to 1. This transfer function is commonly used in back propagation networks, in part because it is differentiable.

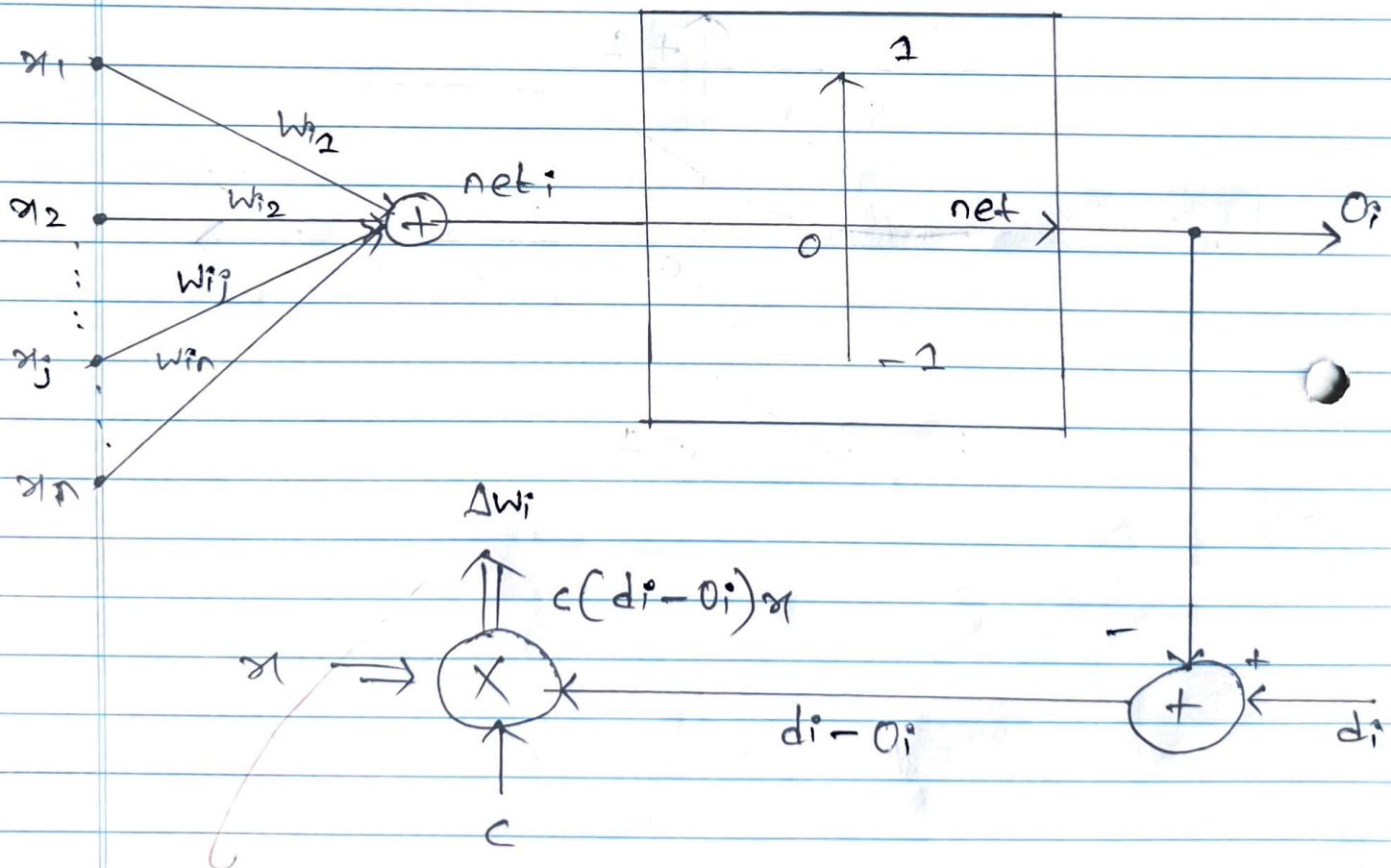
$$F(x) = \frac{1}{1 + e^{-xneL}}$$



Q.2 Explain discrete perceptron learning rule with the help of neat feed forward neural network architecture diagram.

⇒

- The perceptron learning algorithm is a supervised learning algorithm for binary classification, typically used in neural networks to train a single-layer perceptron.
- The perceptron aims to find a linear decision boundary that separates two classes.



Working

Given are P training pairs
 $\{x_1, d_1, x_2, d_2, \dots, x_P, d_P\}$ where

$$x_i = (n \times 1),$$

$$d_i = (1 \times 1), \quad i = 1, 2, \dots, P$$

Note that augment input vectors are used:

$$y_i = \begin{bmatrix} x_i \\ 1 \end{bmatrix}, \quad i = 1, 2, \dots, P$$

Step 1: $\epsilon > 0$ is chosen

Step 2: Weights are initiated at w at small random values, w is $(n+1) \times 1$.
 Counters and errors are initialized:

$$k \leftarrow 1, \quad p \leftarrow 1, \quad E \leftarrow 0$$

Step 3: The training cycle begins here.
 Input is presented and output is computed:

$$y \leftarrow y_p, \quad d \leftarrow d_p$$

$$O \leftarrow \text{sgn}(wt \cdot y)$$

Step 4 : Weights are updated.

$$W \leftarrow W + \frac{1}{2} e (d - o) y$$

Step 5 : Cycle error is computed.

$$E \leftarrow \frac{1}{2} (d - o)^2 + E$$

Step 6 : IF $p < P$ then $p \leftarrow p + 1$,
 $K \leftarrow K + 1$ and go to step 3, otherwise
go to step 7.

Step 7 : The training cycle is completed.
For $E = 0$ terminate the training
session.

IF $E > 0$, then $E \leftarrow 0$, $p \leftarrow 1$,
and enter the new training cycle by
going to step 3.

Q. 3 Apply discrete perceptron learning rule to obtain weights after 2 iteration for the following data patterns. Assume learning rate $c = 0.1$. Use binary bipolar activation function.

$$X_1 = [1 \ -2 \ 0 \ 1]^T, X_2 = [0 \ 1.5 \ -0.5 \ -1]^T, \\ X_3 = [-1 \ 1 \ 0.5 \ -1]^T, d_1 = -1, d_2 = -1, d_3 = 1$$

⇒ Soln:-

$$W_1 = [1 \ -1 \ 0 \ 0.5]^T$$

For bipolar binary output = -1, if net < 0
= 1, if net ≥ 0

⇒ When X_1 is applied:

$$net_1 = W_1^T * X_1$$

$$= [1 \ -1 \ 0 \ 0.5] \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix} = 2.5$$

$$O_1 = f(net_1) = 1, \text{ as } d_1 \neq O_1$$

$$\therefore W_2 = W_1 + c * (d_1 - O_1) * X_1$$

$$\therefore W_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0.5 \end{bmatrix} + (0.1)(-1-1) \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.8 \\ -0.6 \\ 0 \\ 0.7 \end{bmatrix}$$

2) When X_2 is applied:

$$net_2 = W_2^t * X_2$$

$$= \begin{bmatrix} 0.8 & -0.6 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix} = -1.6$$

$$o_2 = f(net_2) = -1 \quad ; \text{ as } net < 0$$

since, $d_2 = o_2$, weight updation is not required.

$$\therefore \text{So, } W_3 = W_2$$

3) When X_3 is applied:

$$net_3 = W_3^t * X_3$$

$$= \begin{bmatrix} 0.8 & -0.6 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix} = -2.1$$

$$O_3 = F(\text{net}_3) = -1, \quad \text{as } d_3 \neq O_3$$

$$W_4 = W_3 + C * (d_3 - O_3) * X_3$$

$$= \begin{bmatrix} 0.8 \\ -0.6 \\ 0 \\ 0.7 \end{bmatrix} + (1)(1 - (-1)) \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.6 \\ -0.4 \\ 0.1 \\ 0.5 \end{bmatrix}$$

(A) ~~9/4/25~~